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The Completion

****A relativistic field theory carrying $a_0 = \kappa c \sqrt{(G\rho - \Lambda)}$ ****

Zimmerman, Carl P. — Briar Creek Tech Assembled 2026-08-09; **v2 same day; v3 same day**. Every numbered property below is backed by a committed, runnable script that exits non-zero on failure. Scripts named at each line. **v2 changes:** row 13’s shift-charge cluster route is **withdrawn** (killed by the 1-Mpc confrontation: smooth accretion restores \$ $\rightarrow 1$ for any cold IC) and replaced by the $**a_{\{0\}} - \text{bump environment response} *$, which passed a five–environment matrix AND its perturbation health check; the health check carved two hard constraints $\leq 8.4 \times 10^{-7}$; amplitude $\leq 2.7 \times$ fiducial). The coefficient row now reads as **measured:** $\kappa = 0.551 \pm 0.043, \pm 0.063$ if the H_0 tension is carried as a systematic ($\kappa \propto a_0/H_0$). **v3 changes:** (i) the **full AeST perturbation matrix** for the bump term is now **DONE** and added as row 17 — $c_T = 1$ exact, no-ghost as a theorem, quasi-static closure derived, amplitude cap softened to a band; non-claim 2c is updated accordingly. (ii) **Non-claim 2 is corrected:** v2’s “by item 13 it is absent where rotation curves are measured” was a dangling pointer — row 13 was withdrawn *in the same version* — so the galaxy-interior status of the Q-sector dust is **REOPENED** and now stated as its own open problem (non-claim 2d). v2 carried this internal inconsistency for several hours; the correction runs against the framework’s interest and is flagged rather than hidden. (iii) §3’s amplitude cap now states its units ($4.5 \text{ Mpc}^{-2} = 2.7 \times$ the fiducial 1.65 Mpc^{-2}).

1. The action

$$S = \int d^4x \sqrt{-g} \left[\underbrace{\frac{R - 2\Lambda_{\text{bare}}}{16\pi G}}_{\text{Einstein-Hilbert}} + \underbrace{\mathcal{L}_{\text{aether}}[A^\mu, g]}_{\text{unit-timelike } A^\mu A_\mu = -1} + \underbrace{\mathcal{F}(Y, Q)}_{\text{the dark sector}} \right] + S_{\text{matter}}[g, \psi]$$

with the two scalar invariants built from the khronon φ and the aether A^μ ,

$$Q \equiv A^\mu \nabla_\mu \varphi, \quad Y \equiv (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \varphi \nabla_\nu \varphi.$$

This is **Aether-Scalar-Tensor** (Skordis & Złośnik 2021, PRL **127** 161302) — chosen because it is the only relativistic MOND-class theory that reproduces the CMB. The framework’s content is entirely in the choice of the free function , which **factorises**:

$$\boxed{\mathcal{F}(Y, Q) = \underbrace{\frac{a_0^2}{8\pi G} \mathcal{F}_Y\left(\frac{Y}{a_0^2}\right)}_{\text{MOND}} + \underbrace{K(Q)}_{\text{dark energy + dark matter}} + \underbrace{A B\left(\frac{Y}{a_0^2}\right) (Q - Q_0)^2}_{\text{a}_0\text{-bump response (v2)}}$$

$$B(y) = \frac{y}{(1+y)^2}$$

The third term (v2) is a **position-dependent Helmholtz mass** $\mu_{\text{eff}}^2 = A B(g^2/a_0^2)$ **peaked at the framework's own** a_0 — the location costs nothing (Y/a_0^2 is already the Y-sector's normalisation); one new calibrated amplitude $A \approx 1.7 \text{ Mpc}^{-2}$. *The dark sector's response is resonant at the MOND transition, and clusters are the cosmic objects that live at a_0 (R_{500} at $0.33\text{--}0.58 a_0$).*

1.1 The Y-sector — MOND

$_Y$ is fixed by the framework's Route A kernel $\nu(y) = 1/(1 - e^{\widehat{(-y)}})$, $y = g_bar/a_0$. In AQUAL variables it is the closed parametric pair

$$\mu(u) = 1 - e^{-u}, \quad x(u) = \frac{u^2}{\mu(u)}, \quad u = \sqrt{y}, \quad x = g_{\text{obs}}/a_0.$$

with the normalisation

$$a_0 = \kappa c \sqrt{G \rho_\Lambda} = \frac{c H_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = 9.3619 \times 10^{-11} \text{ m s}^{-2}, \quad \kappa = \frac{1}{2}, \quad Z = 2\sqrt{8\pi/3}.$$

$\kappa = \frac{1}{2}$ is **FITTED, not derived**. See §4.

1.2 The Q-sector — an offset DBI khronon

$$\boxed{K(Q) = -M^4 + \mu^2 \Lambda_D^2 \left[1 - \sqrt{1 - \frac{u^2}{\Lambda_D^2}} \right], \quad u \equiv Q - Q_0, \quad |u| < \Lambda_D}$$

Three jobs from one function:

piece	limit	what it is
the offset $-M^4$	$u = 0, K = 0$	$\rho = -K = M^4, p = K = -M^4$ $\Rightarrow \mathbf{w} = \mathbf{-1}$ exactly : dark energy
the deviations	$u \ll \Lambda_D$	$K \rightarrow \mu^2 u^2/2 \Rightarrow w \rightarrow 0$: dust , i.e. the dark matter
the saturation	$u \rightarrow \Lambda_D$	K bounded, $K \rightarrow \infty \Rightarrow w \rightarrow 0$ again: kills the early stiff phase

Parameters: M (vacuum scale, $M^4 = \rho \Lambda$), μ (Helmholtz mass), Λ_D (DBI field-space scale), I_0 (the conserved shift charge = the dust amount).

2. What is verified

#	property	value / statement	script
1	Lensing — the test that killed modified inertia	$\Phi = \Psi \implies \gamma_{\text{PPN}} = 1$, $M_{\text{dyn}}/M_{\text{lens}} = 1$ exactly. $21.2\sigma \rightarrow 0.601\sigma$	mi_relativistic_completion_aest.
2	Kernel embeds in $_{\text{Y}}$	deep-MOND $\mu \rightarrow x$ exact; Newtonian $\$ \rightarrow 1; x(u) \text{bijection}(h = e^{(-u)}(1+u) > 0); \text{free function} * \text{convex} * same 3 * \text{Newtonian residual} * e^{(-\sqrt{\$y})} = 3.6 \times 10^{-3457}$ at Earth's orbit	same
4	Dark energy is not an input	$w = -1$ exactly at the condensate minimum — proved, not assumed	mi_condensate_vacuum_energy_a0_2
5	Ghost-free over the whole field range	$K = \mu^2(1-s^2)^{-3/2} > 0$ for all $ s < 1$	mi_dbi_khronon_2026.py
6	Subluminal everywhere	$c_s^2 = \Lambda_D s(1-s^2)/(1+\Lambda_D s) \leq 0.385 \Lambda_D$	same
7	The published no-go is dissolved	Blanchet & Skordis 2024 §4.3.1's 455× conflict reverses sign : bounded pressure $\implies w \rightarrow 0$ early $\implies$ cosmology becomes a <i>lower</i> bound on μ^{-1} , same direction as MOND	same
8	Theorem : why no polynomial works	for $K \sim u^n$, early-time $w \rightarrow 1/(n-1)$. Quadratic 1, quartic $\frac{1}{3}$, sextic — all fail. Only boundedness gives $w \rightarrow 0$	same

#	property	value / statement	script
9	CMB acoustic peaks — real CLASS run	at $\Lambda_D \leq 10^{-2}$, $c_s^2(\text{recomb}) = 2.9 \times 10^{-8}$; TT unchanged to 0.069% , $P(k=0.2)$ to 1.71%. Indistinguishable from CDM	mi_dbi_cmb_class_run_2026.py
10	BTFR	exact, from convexity	mi_route_a_field_theory_2026.py
11	g^{-2} Lorentz violation	restored by the aether (pure Bekenstein–Milgrom had lost it)	mi_relativistic_completion_aest.
12	a_0 tied to Λ	$a_0 = m_{\text{cond}}/(4\sqrt{\pi})$ with $m_{\text{cond}} = M^2/(\sqrt{2}M_{Pl}), M^{\{4\}} = \rho_{\Lambda}$ — agrees to 0.076%	mi_condensate_vacuum_energy_a0_2
13	Clusters (v2)	shift-charge IC route withdrawn: killed by the 1-Mpc confrontation — cluster R500 and galaxy-outskirt lensing share a scale with disjoint requirements, correlated Gaussian ICs cannot select environments, and smooth accretion restores $\$ \rightarrow 1 \text{ for any cold IC} *$ $*. \text{Replaced by the} *$ $*a\{0\}\text{--bump response} *$ $*(\text{row } 15) \backslash \text{mi}_{i,c,\text{route } 1 \text{ mpc into function } 2026.py} (9/9) 14 *$ $*\text{Directional EFE signal} *$ $* \text{pure MI predicted} *$ $\text{exactly zero} *$; AQUAL — $\text{class predicts } 1\text{--}4 15 *$ $*a\{0\}\text{--bump environment response}$	passes all five environments: cluster (calibration, $\mu^2_{\text{eff}} = 0.23 \text{ Mpc}^{-2}$), galaxy interior 0.034% of M_b (RAR cost 4×10^{-4} dex), galaxy 1 Mpc 1.2% vs the 5.9% strict bound, linear cosmos 0.6% of mean matter, solar 10^{-19} . Vanishes on FRW ($Y=0$) and is second-order in perturbations $\Rightarrow$ linear CMB/ $P(k)$ construction. Evades both prior kills (a response has no charge to advect; couples to environment directly)

#	property	value / statement	script
16	...and its health check	no ghosts ever (kinetic contribution $2AB \geq 0$, $= 0$ on FRW), no Ostrogradsky sector; FRW gradient health **excludes the old $\Lambda_D = 10^{-2}$ and forces $\Lambda_D \leq$ 8.4×10^{-7} ** (window stays 3.7 orders); halo gradient health caps $A \leq 2.7 \times \text{fiducial}$, excluding the demanding end of Mistele's cluster band — a two-sided falsifiable pinch; free signature: cluster anisotropic stress at $O(0.6)$	mi_a0_bump_health_2026.py (11/11)

#	property	value / statement	script
17	...and the FULL perturbation matrix (v3)	with all aether mixings: c_T = 1 exact (the new terms are algebraic in the metric — GW170817 safe by construction); no-ghost is a THEOREM (the bump’s kinetic addition is rank-1 PSD in the $\chi\chi$ entry $\Rightarrow$ Weyl monotonicity: it cannot lower any eigenvalue of AeST’s healthy kinetic matrix; closed-form $2\times 2 +$ 200 random 4×4); unit-norm gives $\delta A^0 =$ $-\Phi A^0 \Rightarrow \delta Q = \chi -$ $Q_0\Phi$ — the quasi-static phenomenology of row 15 is derived, not assumed ; gradient block exact: $\Delta G =$ $2\lambda[[q,$ $2y^{\{3/2\}}B],[2y^{\{3/2\}}B,$ $yq]], q =$ $(3y^2-8y+1)/(1+y)^4$; integrating out the aether softens the amplitude cap: at $base_a = 1$ the band is A_max $\in [2.72,$ 4.46] $\times$ fiducial $=$ [4.49, 7.36] Mpc^{-2} (v4 correction : the earlier “(2.7–7.4) $\times$ fiducial” spliced fiducial units with Mpc^{-2} , making the band look $1.66\times$ wider at the top than it is — <code>base_a_lookup_and_robustness_2026.py</code> , 12/12). The base_a lookup is now DONE and it 6 converts the caveat rather than closing it : <code>base_a</code> is built from AeST’s single	<code>mi_aest_full_matrix_bump_2026.py</code> (11/11)

3. Parameter count, honestly

	Λ CDM	this completion
dark sector	Ω_c, Ω_Λ	$M (= \rho_- \Lambda^{\{1/4\}}), I_0, \mu, \Lambda_D, A$ (v2)
MOND scale	—	a_0 — not independent , = $m_{\text{cond}}/(4\sqrt{\pi})$ via item 12
bounds	—	$\Lambda_D: 1.9 \times 10^{-10} \ll \Lambda_D \leq 8.4 \times 10^{-7}$ (health-bounded, v2 ; 3.7 orders); $A \approx 1.7$ Mpc^{-2} calibrated on clusters, health cap $A \leq 4.5 \text{ Mpc}^{-2} = 2.7 \times \text{fiducial}$ (row 16; the full matrix widens it to $[2.72, 4.46] \times \text{fiducial} = [4.49, 7.36] \text{ Mpc}^{-2}$ at $\text{base}_a = 1$ — row 17, where the base_a fork and its robustness are stated); $\mu^{-1} = 4392 \text{ Mpc}$ (item 12); $I_0 \approx \Omega_{\text{dm}}$ (an IC)

So: **five dark-sector numbers against Λ CDM's two** (v2 adds A), with a_0 derived from one of them rather than added. Not fewer parameters. What it buys is that Λ , dark matter and MOND come from **one function** instead of three unrelated sectors.

4. What is NOT claimed — read this before quoting anything above

1. $\kappa = \frac{1}{2}$ **IS NOT DERIVED**. Item 12 looks like a derivation and is not. $a = m_{\text{cond}}/(4\sqrt{\pi})$ is *algebraically identical* to $a = \sqrt[4]{G_\Lambda}$ and to $Z = \sqrt[4]{(8/3)}$ — three costumes, one statement — and the residue is **convention-dependent** ($4\sqrt{\pi}$ with the reduced Planck mass, $\sqrt[4]{2}$ with the non-reduced). A derived number cannot depend on a bookkeeping choice. It is a **relabelling**, as this corpus's own theorem already established. What genuinely moved: “why $\kappa = \frac{1}{2}$?” becomes “why is $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$?”, which is sharper because m_{cond} is a *derived* quantity of a real field theory rather than a definition.
2. **Dark matter EXISTS**, at the full Ω_{dm} . Removing the pressureless component moves H_3/H_1 by 54% and no refit absorbs it ($\Delta\chi^2 > 400$). What does **not** exist is a dark-matter *particle*: it is the Q-sector of the same scalar that supplies Λ and MOND. (**v3 correction**: v2 continued “and by item 13 it is absent where rotation curves are measured” — a dangling pointer, since v2 itself withdrew row 13. See 2d. **The defensible slogan shrinks to “no dark-matter particle” — never “no dark matter”, and “none in galaxies” is now an open problem, not a result.**) 2d. (**v3**) **WHERE THE DUST SETTLES IS THE SHARPEST OPEN PROBLEM**. The 1-Mpc confrontation's own smooth-accretion

theorem (row 13's killer) implies the Q-sector dust falls into *every* collapsing basin, galaxies included. Two branches are then live: **(favorable)** the settled condensate relaxes to the Helmholtz-preferred profile — ρ_c tracking the potential, flat, centrally evacuated (`mi_virialisation_verdict_2026.py`: $\xi \propto R^2$, galaxy-interior cost $\sim 6 \times 10^{-5}$ dex) — and galaxies are safe by orders of magnitude; **(fatal)** the dust retains CDM-like central concentration and the completion overshoots its own best data by the banked $2.06\text{--}4.42\times$. Which branch obtains is a **nonlinear dynamics question — relaxation of the condensate toward the static profile within a Hubble time — that no group has ever computed for AeST**. It is scheduled as this program's next calculation (N-body-class, staged from spherical symmetry up). Until it is done, galaxy-interior cleanliness is an *assumption the theory needs*, not a result it owns. **(Same-day postscripts, stages 1 AND 2.** Stage 1 (`nbody_2026/stage1_condensate_relaxation_2026.py`, 11/11) found a sound-crossing bound $\Lambda_D \gtrsim 1.2 \times 10^{-9}$. **Stage 2 (`nbody_2026/stage2_spherical_collapse_2026.py`, 10/10) WITHDRAWS stage 1's favorable branch — crossing ability is necessary, not sufficient — and finds the fatal branch is the DEFAULT at every Λ_D in the health window, within the fluid description:** the bounded DBI pressure cannot support galactic dust (even the optimal $c_s^2 = 0.385\Lambda_D$ at the ceiling falls $1.38\times$ short of the halo virial parameter), the Helmholtz profile's capacity at $\mu^{-1} = 4392$ Mpc is $\sim 10^{-7}$ of the captured charge, and the basin's dust free-falls to a sub-kpc caustic within a Hubble time — overshooting the 10-kpc RAR point by ~ 0.72 dex. **Stage 2b** (`nbody_2026/stage2b_base_a_attribution_and_A_conflict_2026.py`, 15/15) then corrected TWO of stage 2's own claims, in opposite directions. (i) *A hope withdrawn*: stage 2 said `base_a` could move the FRW ceiling and so carried “mortal weight” for the galaxy problem — **false**; the ceiling is $\sqrt{(\text{CLASS_DMG} \cdot \mu^2 / (2A \cdot L_0^2 \cdot \text{pk}))}$, in which `base_a` does not appear, and its only possible channel was already bounded at 3.2×10^{-4} by row 17's F1. `base_a`'s real and only load is the **cluster** pinch (the Mistele $34\times$ end stays excluded for any `base_a` $\in [0.5, 4]$; the $4\times$ edge is marginal for `base_a` $\lesssim 1$). (ii) *A deficit withdrawn*: stage 2's “supported $\Rightarrow$ possibly survivable” was **right**, and stage 2b's attempt to tighten it rested on a circular estimator (caught by its own negative control); the honest figure is 0.052 dex at 1 Mpc, inside the ~ 0.1 dex outskirts tolerance. **What replaces the lost door is tighter than it was**: the ceiling scales as $A^{(-1/2)}$, so galactic dust support needs $A \leq 0.87 \text{ Mpc}^{-2}$ while clusters calibrate $A = 1.65$ and Ref. Mistele's modelling wants 6.6–56 — **one parameter, two jobs, opposite directions; the bump's success on clusters monotonically deepens its galaxy failure**. And a **new falsification on existing data**: support is v_c -dependent, splitting at $v_c = c \sqrt{(0.385 \Lambda_{D,\text{max}})} = 170 \text{ km/s}$ — *below it the interior is clean, above it spirals sit 0.72 dex off the RAR* — so the fluid branch predicts a break inside the SPARC range that **is not observed**. Remaining door: stage 3 (wave/field dynamics at the caustic, outside the fluid description), which must keep interiors clean at $v_c > 170 \text{ km/s}$ without lowering A . 2d remains OPEN; its fluid-description branch is now falsified rather than merely disfavoured.) **2b. The coefficient is MEASURED, not derived**: $\kappa = 0.551 \pm 0.043^{**}$ (distance-free SPARC estimator), and ± 0.063 if the H_0 tension is carried as a systematic, since $\kappa \propto a_0/H_0$ at full strength. $\frac{1}{2}$, $1/\sqrt{3}$, $\sqrt{(3/8)}$ and 0.40 all sit inside 2σ . (*v2 correction: v1's ± 0.043 omitted the $\rho\Lambda$ term.*) 2c. **The a_0 -bump response is a CANDIDATE, not a result**: it has passed the environment matrix, its isolated-term health check, and (v3) the full AeST perturbation matrix with aether mixings (row 17). Still owed: a **real cluster-model fit** (the amplitude is calibrated on a single summary number — nobody,

this program included, has fit real cluster mass profiles), the **base_a identification** — the *lookup* is done (row 17: K_B is a never-measured free parameter bounded to $0 < K_B < 2$), but which K_B -combination the mixing entry equals is a short derivation still owed; it decides only the $4\times$ edge, the $34\times$ exclusion being robust — and a dedicated vector-sector analysis. The health-vs-Mistele amplitude pinch could still kill it.

3. **The post-recombination growth history is NOT verified.** Item 9 covers the acoustic peaks only. Holding c_s^2 at its peak value for all time *destroys* $P(k=0.2)$, so the pass depends essentially on the bump being transient (it peaks at $z \approx 189$). **A patched CLASS fluid carrying $c_s^2(a)$ is required, not optional.**
4. **(v4 rewording — the original referred to withdrawn row 13.)** The dust component's small-scale initial conditions are **not confronted with Lyman- α** . The withdrawn IC route needed a khronon transfer function $T \approx 0.33$ at $k \approx 4.5 \text{ Mpc}^{-1}$ (Λ CDM-excluded); the surviving picture (standard cold dust + the bump response) should be Ly- α -safe by construction, but that is **asserted, not computed — the confrontation remains owed item in the programme.**
5. **Cassini Q_2 is inherited** ($3\text{--}15\sigma$) and is not relieved by anything here.
6. **PPN preferred-frame parameters α_1, α_2** for this have not been computed against lunar-laser and binary-pulsar bounds.
7. **The wide-binary target is the point-field asymptote only.** $\gamma_v = \sqrt{\nu(y_{\text{extN}})}$; the full nonlinear AQUAL-EFE solve is owed.
8. **This is not a theory of everything** and no part of it addresses the Standard Model. The 2026-06-23 retraction stands.

5. The falsifiable predictions

prediction	value	how to test
wide-binary boost	$\gamma_v = \sqrt{\nu(y_{\text{extN}})}$	Gaia DR4 , pre-registered and hash-stamped
directional EFE asymmetry	1–4%, signed	already firing at $p = 0.029$ with the right sign
cluster-to-cluster scatter in the residual	$\sigma(\xi)/\xi \approx 0.02\text{--}0.60$	existing cluster samples — (v4) separates the a_0-bump response (scatter tracks each cluster's distance from the resonance, rows 15–17) from any fixed-scale mechanism, which predicts uniformity (the two prior candidates are withdrawn)
cluster mass profile misfit	ρ_c flat, $M_c \sim r^3$, can go negative	fit clusters with the khronon profile instead of NFW
g^{-2} Lorentz violation	computable $s_{\mu\nu}$	SME sector bounds
accelerated structure formation	earlier massive objects	JWST high- z massive galaxies (McGaugh et al. 2024)

6. Credit

$\nu = \sqrt{1+1/y}$ is **Milgrom 1999 PLA 253:273 eq 9**. MOND: Milgrom 1983 ApJ 270:365. AQUAL: Bekenstein & Milgrom 1984 ApJ 286:7. TeVeS: Bekenstein 2004 PRD 70:083509. **AeST: Skordis & Złośnik 2021 PRL 127:161302** — the completion itself. Ghost condensate: Arkani-Hamed, Cheng, Luty & Mukohyama 2004 JHEP 0405:074; nonlinear: +Wiseman 2007 JHEP 0701:036. $K(Q) = \mu^2(Q-1)^2$ and the ghost- condensate identification: Verwayen, Skordis & Złośnik 2024; **Blanchet & Skordis 2024 JCAP 11(2024)040** — including §4.3.1, the no-go this work dissolves. Quasi-static AeST: Durakovic & Skordis 2024 JCAP 04:040; Verwayen, Skordis & Boehm 2024 MNRAS 531:272. **Mistele, McGaugh & Hossenfelder 2023 A&A 676:A100** — the weak-lensing bounds that killed this program’s first two cluster mechanisms and now two-sidedly bound the third (rows 15–17). GDM bound: Kopp, Skordis, Thomas & Ilić 2018 PRL 120:221102. Ly- α floor: Rogers & Peiris 2021 PRL 126:071302. RAR scatter: Desmond 2023. Accelerated structure formation: McGaugh, Schombert, Lelli & Franck 2024.

The distinctive content of this work is the coefficient — $a_0 = \kappa c \sqrt{(G\rho_Y)}$ with $\kappa = \frac{1}{2}$ — its embedding in $_Y$, the offset-DBI Q-sector, and items 7, 8, 9, 12 and 13. Everything else is cited above and is not mine.